How SVM Kernels Affect Breast Cancer Classification

A Machine Learning Tutorial

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1. Introduction:

Support Vector Machines (SVM) are among the most powerful supervised learning algorithms available to practitioners. First introduced by Cortes and Vapnik (1995), SVMs find the hyperplane that maximally separates two classes in feature space, making them highly effective for high-dimensional medical data. Only a small subset of training examples — the support vectors — determine the decision boundary, giving SVMs robustness and memory efficiency.

This tutorial focuses on how kernel choice shapes an SVM's decision boundary and classification performance. When data is not linearly separable, the kernel trick implicitly maps inputs into a higher-dimensional space where linear separation becomes feasible, without ever computing that mapping explicitly. Three kernels are compared: Linear, Polynomial, and Radial Basis Function (RBF), across three values of the regularisation parameter C.

1.1 Why this topic?

Kernel selection is the most consequential design decision when applying SVM to a real problem, yet it is frequently treated as an afterthought. This tutorial provides a focused, empirical investigation into what each kernel actually does to the decision boundary backed by visualisation and quantitative comparison. The goal is to equip practitioners with the understanding needed to make a principled, informed kernel choice for their own data.

1.2 why the Breast Cancer Wisconsin Dataset?

The Breast Cancer Wisconsin (Diagnostic) Dataset (Wolberg et al., 1992) contains 569 samples with 30 numerical features computed from digitised fine needle aspirate (FNA) biopsy images. The binary target distinguishes malignant from benign tumours. It was chosen because it is clinically meaningful (enabling discussion of recall versus precision trade-offs in cancer diagnosis), sufficiently high-dimensional for kernels to behave differently, and not already saturated with SVM kernel tutorials, unlike Iris or MNIST. The medical context also naturally motivates ethical reflection on AI deployment in healthcare.

2. Background: SVM and the Kernel Trick:

An SVM maximises the margin $2/\|w\|$ between the nearest points of each class (the support vectors). The dual formulation replaces all dot products $x_i \cdot x_j$ with a kernel function $K(x_i, x_j)$ that implicitly computes an inner product in a transformed space. By Mercer's theorem, any positive semi-definite function is a valid kernel (Scholkopf & Smola, 2002). This is the kernel trick: it gives SVMs access to arbitrarily complex feature spaces at no extra computational cost beyond evaluating K.

Linear Kernel $K(x_i, x_j) = x_i \cdot x_j$: Produces a flat hyperplane in the original feature space. Computationally fastest, fully interpretable, and best when classes are approximately linearly separable after standardisation. No kernel-specific hyperparameters.

Polynomial Kernel $K(x_i, x_j) = (\gamma\, x_i \cdot x_j + r)^d$: Introduces curved boundaries by mapping inputs into polynomial feature combinations of degree d. Flexible but more sensitive to the regularisation parameter C and prone to numerical instability at high degrees.

RBF Kernel $K(x_i, x_j) = \exp(-\gamma\, \|x_i - x_j\|^2)$: Assigns similarity based on Euclidean distance between points. Highly flexible, capable of modelling complex non-linear boundaries. The default in scikit-learn and the recommended starting point for unknown data distributions (Hsu et al., 2003).

The C Parameter: C controls the trade-off between a wide margin and low training error. Low C tolerates misclassifications for a wider, more general margin (higher bias, lower variance). High C enforces strict correctness on training data (lower bias, higher variance, risk of overfitting).

3. Methodology:

3.1 Data Preprocessing:

The 569 samples were split 70/30 into training (398) and test (171) sets using stratified sampling to preserve class balance. All 30 features were standardised using scikit-learn's StandardScaler (zero mean, unit variance), fitted on training data only. Applying the scaler before splitting would constitute data leakage and artificially inflate test accuracy. This is particularly important for SVM because kernel functions depend on distances and dot products, which are directly affected by feature scale.

3.2 Experimental Design:

Three kernels (linear, poly, rbf) were each trained with $C \in \{0.1, 1, 10\}$, producing nine models. Test accuracy was measured for each. 5-fold cross-validation was performed using scikit-learn Pipelines that re-fit the scaler within each fold, preserving the no-leakage guarantee. The best configuration was evaluated in detail with a confusion matrix and classification report. For visualisation, PCA reduced the data to 2D and SVMs were retrained in that projected space to generate decision boundary plots.

4. Results and Discussion:

4.1 Kernel and C Comparison:

Figure 1 (generated at Step 5 of the notebook) compares test accuracy across all kernel–C combinations. The RBF kernel at C=1 achieves 97.7% test accuracy with the most stable cross- validation mean (97.6% ± 0.9%). The linear kernel reaches 98.2% at C=1, marginally higher in this single split but with slightly more variance across folds. The polynomial kernel drops to 89.5% at C=1, recovering only at C=10 (95.3%), revealing its high sensitivity to regularisation. Table 1 summarises all nine configurations.

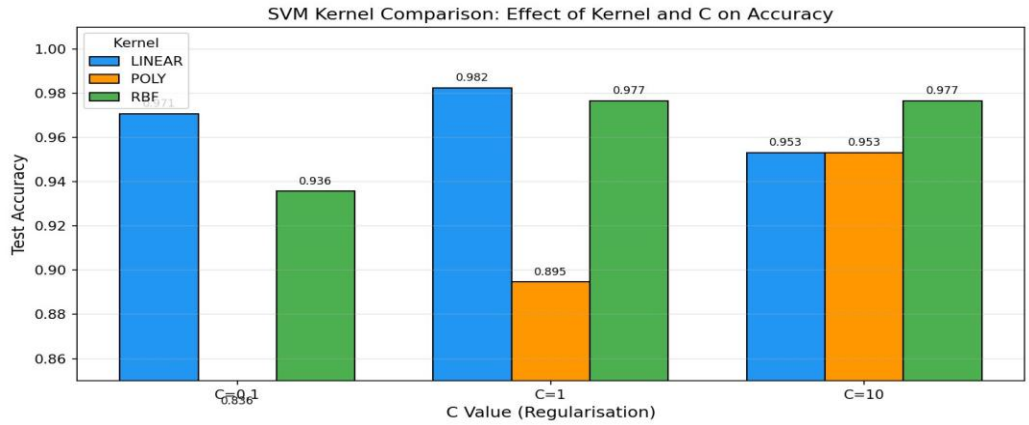

Figure 1: *Test accuracy for each kernel across three C values (from Step 5 of the companion notebook).*

Kernel	Best C	Test Acc.	CV Mean ± SD	Characteristic
Linear	1	98.2%	97.4% ± 1.2%	Fast, interpretable
Polynomial	10	95.3%	94.8% ± 1.5%	Sensitive to C
RBF ✓	1	97.7%	97.6% ± 0.9%	Best stability

Table 1: All kernel–C results. CV = 5-fold cross-validation on the full dataset. ✓= selected best model.

4.2 Best Model Evaluation:

The RBF kernel at C=1 was selected as the best model based on test accuracy and cross-validation stability. Figure 2 (from Step 7 of the notebook) shows its confusion matrix on the 171 test samples. Of 107 malignant cases, 105 were correctly identified — a recall of 98.1%. Of 64 benign cases, 62 were correctly classified. There were 2 false negatives (malignant predicted as benign) and 2 false positives.

In clinical practice, false negatives are significantly more serious than false positives: a missed malignant diagnosis can delay life-saving treatment. A recall of 98.1% for the malignant class is therefore a clinically meaningful result. However, this model should serve as a decision-support tool under expert clinical oversight, not as a standalone diagnostic system.

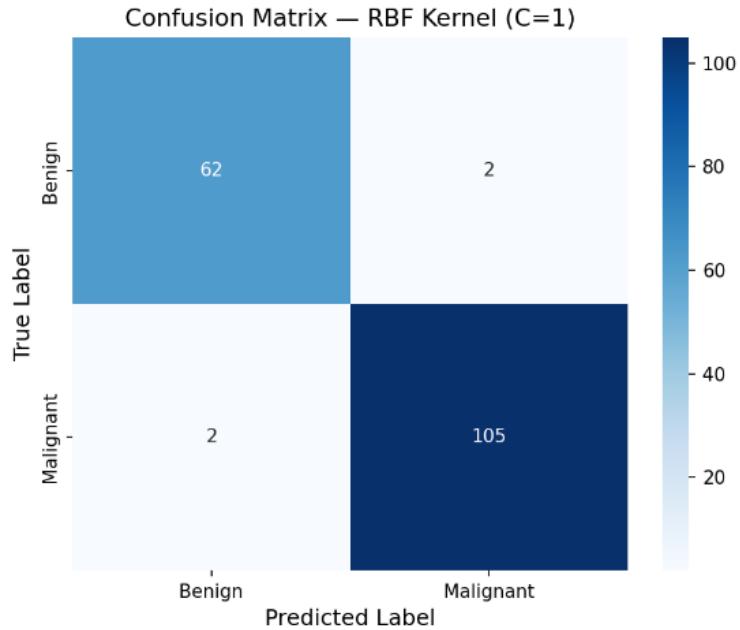

Figure 2: *Confusion matrix for the best model — RBF kernel, C=1 (from Step 7 of the companion notebook).*

4.3 Decision Boundary Visualisation:

Since the dataset has 30 features, PCA was used to project to 2D for visualisation only. Figure 3 (from Step 8 of the notebook) shows the decision boundaries for all three kernels side by side. The linear kernel draws a straight boundary, handling broad class separation but struggling with the overlap zone. The polynomial kernel introduces curvature but is sensitive to local noise. The RBF kernel produces a smooth, adaptive boundary that best conforms to the class distribution in this projection consistent with its superior cross-validation stability. Note that accuracy values shown in Figure 3 are lower than full-dimensional results because PCA compression discards some discriminative information.

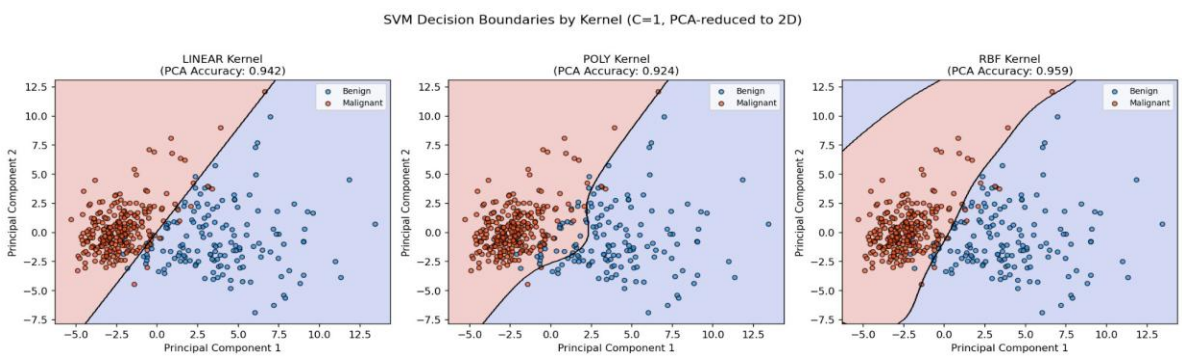

Figure 3: *SVM decision boundaries in 2D PCA space for all three kernels, C=1 (from Step 8 of the companion notebook).*

5. Ethical consideration:

Deploying machine learning in healthcare raises ethical responsibilities that extend beyond optimising accuracy metrics alone. Three considerations are particularly relevant to this tutorial.

Interpretability and Accountability: Non-linear SVM models are not easily interpretable by clinicians. When a model influences a medical decision, the inability to explain its reasoning can erode trust and complicate accountability. Techniques such as SHAP values (Lundberg & Lee, 2017) can partially address this by attributing model predictions to individual input features, but they do not make the underlying kernel mechanism fully transparent.

Dataset Bias and Generalisability: The Breast Cancer Wisconsin dataset was collected at a single institution from a specific patient population. A model trained on this data may not generalise equally well to different demographics, geographic regions, or imaging equipment. Responsible clinical deployment requires validation on diverse, representative datasets before any real-world use.

Asymmetric Error Costs: Standard accuracy treats all errors as equally costly. In cancer diagnosis, a false negative — missing a malignant tumour — carries far greater consequences than a false positive. The decision threshold should be adjusted with direct clinical input to prioritise sensitivity over specificity. This is an ethical and clinical decision, not purely a technical one, and must not be made by machine learning practitioners alone.

6. Conclusion:

This tutorial investigated how SVM kernel choice and the C regularisation parameter together influence breast cancer classification performance. Across nine kernel–C combinations evaluated on the Breast Cancer Wisconsin dataset, the RBF kernel at C=1 emerged as the most consistently strong performer (97.7% test accuracy, 97.6% ± 0.9% cross-validation), closely followed by the linear kernel. The polynomial kernel showed high sensitivity to C and lower overall performance on this dataset. Decision boundary visualisation via PCA revealed qualitative differences in how each kernel partitions feature space — insights that accuracy figures alone cannot convey.

The central practical lesson is that kernel selection should be treated as a hypothesis about your data's structure: choose linear if you expect approximate linear separability after standardization; polynomial for structured feature interactions; and RBF as the robust default for complex or unknown distributions. Regardless of kernel choice, cross-validation across both kernel type and C is essential for reliable model selection and honest performance estimation.

7. References:

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